

Arthus Goujon

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Education

École Polytechnique – Institut Polytechnique de Paris

Sep 2026 – June 2027

Master in Economics

- Relevant coursework: Econometrics, Time Series, Game Theory, Macroeconomics
- Auditing selected graduate-level courses from Sorbonne University's MS2A Master's program, including Statistical Learning and Optimization for Machine Learning.

Aix-Marseille Université

Sep 2025 – June 2026

M1 Mathematics and Applications

- Relevant coursework: Probability and Measure Theory, Stochastic Processes, Partial Differential Equations, Analysis, Algebra
- Research seminar: Amenable Groups

HEC Paris

Sep 2023 – May 2027

Master in Management, “Grande École” Program

- Completed a Bachelor's degree in Mathematics in parallel with the HEC curriculum.
- Relevant coursework: Data Science, Statistics, Macroeconomics, Microeconomics

Institution des Chartreux

Sep 2020 – May 2023

Classes Préparatoires – Mathematics & Economics

Research Experience

CREST – Télécom Paris – Paris, France

Sep 2026 – Present

Research Project – Quantitative Finance & Market Microstructure

Advisor: *Michele Fabi*

- Study mean-field game models of limit order book dynamics to characterize strategic interactions, equilibrium behavior and liquidity formation among heterogeneous market participants.

CNRS – HEC Paris – Paris, France

March 2026 – August 2026

Research Intern – Applied Mathematics & Game Theory

Advisor: *Marie Laclau*

- Developed a dynamic Bayesian model of strategic scoring under imperfect information and probabilistic auditing; characterized truthful, pooling and interior stationary equilibria.
- Derived incentive and stability thresholds linking score reliability, endogenous exposure and strategic discipline; the reference calibration yields an unstable tipping point around 0.37.

Aix-Marseille Université – Marseille, France

2025 – 2026

Master's Research Thesis – Quantitative Finance & Convex Optimization

Advisor: *Sebastian Müller*

- Formulated optimal routing across CFMM networks as a convex nonlinear flow problem and derived dual no-arbitrage certificates from local pool prices and transaction fees.
- Implemented the framework in Python/CVXPY on a three-pool Uniswap V2 network; optimal order splitting improved execution by up to **22 bps** for a 100 WETH trade.

Machine Learning Project

Mario Party: Tricky Turntable – Game Theory, Bayesian Learning & Reinforcement Learning

- Extended an analytically solved Mario Party game into an adaptive-agent benchmark with **60M+ simulated interactions**, comparing Bayesian opponent modeling, online adaptation and tabular Q-learning.
- State-conditioned Bayesian learning achieved near-oracle performance against reactive opponents; under concept drift, sliding-window inference adapted **5.3× faster** than model-free Q-learning and reduced post-change regret by **89%** versus full-history Bayesian updating.

Technical Skills

Programming – Python (NumPy, pandas, scikit-learn, statsmodels, CVXPY, matplotlib), SQL

Machine Learning – Bayesian learning, reinforcement learning (tabular Q-learning), online learning, statistical learning

Mathematics – Probability, Stochastic Processes, Statistics, Linear Algebra, Convex Optimization, Game Theory

Languages – French (native), English (fluent)